

PAVLODAR, Kazakhstan

WMO: 360030

Lat: 52.2003N

Lon: 77.0703E

Elev: 122

StdP: 99.87

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-31.8	-38.8	0.1	-35.1	-35.1	0.1	-31.6	12.3	-6.8	10.8	-7.9	1.6	0	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	33.0	18.3	31.1	18.0	29.1	17.5	20.5	28.0	19.6	27.1	18.8	26.2	3.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.2	23.0	17.1	12.4	22.5	16.1	11.6	21.7	59.8	28.1	56.8	26.9	54.1	26.1	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.6	9.2	8.0	-36.6	36.7	4.5	1.8	-39.8	37.9	-42.4	39.0	-44.9	40.0	-48.2	41.2	DB WB
			-36.7	22.2	4.4	1.1	-39.9	23.0	-42.5	23.6	-45.0	24.2	-48.2	25.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-17.5	-15.0	-6.0	6.6	13.6	19.5	20.9	19.0	12.4	4.7	-5.5	-13.3
	DBStd	14.91	8.42	7.43	7.17	5.58	5.03	4.05	3.40	3.86	4.70	5.07	7.76	7.65
	HDD10.0	3596	852	701	497	129	23	1	0	1	28	176	466	723
	HDD18.3	5704	1111	934	755	351	160	34	14	39	187	422	716	982
	CDD10.0	1177	0	0	1	28	136	286	339	278	99	12	0	0
	CDD18.3	244	0	0	0	1	14	69	94	59	8	0	0	0
	CDH23.3	2957	0	0	0	23	256	829	985	714	148	2	0	0
	CDH26.7	1077	0	0	0	4	76	314	371	263	49	0	0	0
Wind	WSAvg	3.5	3.3	3.4	3.9	4.0	3.8	3.3	3.1	3.2	3.2	3.4	3.7	3.7
Precipitation	PrecAvg	281	16	13	12	16	26	36	54	30	22	21	16	16
	PrecMax	389	26	25	27	45	108	75	125	74	48	69	29	28
	PrecMin	152	2	3	2	1	7	6	10	9	2	3	7	8
	PrecStd	60	7	6	7	12	22	21	30	19	12	16	6	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	1.9	13.9	26.3	32.5	34.8	35.2	35.1	32.1	22.5	11.3	2.2
		MCWB	0.5	0.4	6.9	13.7	16.9	17.8	19.5	19.1	16.4	11.8	7.1	0.9
	2%	DB	-0.2	0.1	8.9	22.4	28.5	32.2	32.7	32.1	27.3	18.9	8.0	1.0
		MCWB	-1.2	-1.1	4.8	12.0	15.0	17.7	18.9	18.4	14.9	10.4	5.1	-0.2
	5%	DB	-2.8	-1.8	5.1	19.5	26.2	30.2	30.5	29.7	24.1	15.6	5.5	-0.8
		MCWB	-3.9	-2.8	2.7	10.5	14.1	17.6	18.7	17.9	13.6	9.1	3.3	-1.8
	10%	DB	-5.1	-3.9	2.7	16.7	23.7	28.1	28.5	27.3	21.1	12.4	3.5	-2.5
		MCWB	-6.0	-4.9	1.0	9.2	13.5	17.0	18.3	17.0	12.6	7.6	1.8	-3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.6	8.4	14.4	17.9	20.7	22.1	21.1	17.7	13.0	7.9	1.2
		MCDB	1.5	1.4	12.7	23.9	28.5	29.0	29.9	29.7	28.1	19.2	10.4	1.9
	2%	WB	-1.2	-1.0	5.2	12.7	16.6	19.6	20.9	19.8	16.0	11.2	5.2	-0.1
		MCDB	-0.2	0.1	8.6	20.9	25.8	27.8	27.9	28.3	24.8	17.5	7.6	0.8
	5%	WB	-3.7	-2.8	2.9	11.2	15.3	18.7	20.1	18.8	14.6	9.7	3.4	-1.7
		MCDB	-2.8	-1.8	5.1	17.9	23.5	26.8	27.0	27.0	22.3	14.8	5.4	-0.8
	10%	WB	-6.0	-4.8	1.0	9.7	14.2	17.9	19.3	17.8	13.2	8.0	1.8	-3.4
		MCDB	-5.1	-3.9	2.5	15.9	22.0	25.5	26.0	25.9	19.9	12.3	3.4	-2.5
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.3	10.0	12.8	13.5	12.7	11.8	12.9	12.7	10.6	8.2	9.3
		MCDBR	9.4	9.4	11.4	17.4	17.3	16.1	14.9	16.5	17.8	15.8	9.4	9.4
		MCWBR	9.1	8.7	8.0	9.1	7.5	6.4	5.5	6.4	7.9	8.7	7.1	9.0
	5% WB	MCDBR	9.6	9.5	11.0	16.0	15.1	13.3	12.5	14.3	15.6	14.0	8.7	9.5
		MCWBR	9.2	8.9	8.2	8.8	7.3	6.1	5.3	6.1	7.6	8.4	6.9	9.2
Clear-Sky Solar Irradiance	taub	0.304	0.340	0.396	0.419	0.422	0.427	0.451	0.430	0.382	0.355	0.342	0.314	
	taud	2.019	1.942	1.936	2.122	2.168	2.227	2.166	2.199	2.279	2.255	2.123	2.006	
	Ebn at Noon	643	729	767	811	830	828	798	793	790	726	597	546	
	Edh at Noon	82	119	148	139	140	133	139	128	105	87	75	70	
All-Sky Solar Radiation	RadAvg	1.26	2.41	3.77	4.73	6.03	6.39	5.77	5.11	3.72	2.04	1.03	0.82	
	RadStd	0.12	0.14	0.29	0.35	0.43	0.28	0.37	0.31	0.31	0.27	0.11	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page